

PROJECT DOCUMENTATION

Personal Finance Dashboard

Power BI | DAX | Power Query | Star Schema

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1. Project Overview

This project builds an automated Personal Finance Dashboard using Power BI to help individuals track their income, expenses, and savings in a structured and visually intuitive way. Instead of relying on scattered spreadsheets or manual calculations, this dashboard provides a real-time, interactive overview of financial health.

1.1 Problem Statement

Most people do not have a clear picture of where their money goes each month. Manual tracking through spreadsheets is error-prone and time-consuming, making it difficult to identify spending patterns or track progress toward financial goals.

1.2 Objective

- Build an end-to-end automated finance tracker in Power BI
- Answer the core question: Where is my money going, and am I on track to meet my financial goals?
- Provide actionable insights through interactive visuals and KPI metrics

1.3 Key Target Metrics

Metric	Description
Total Income	Sum of all income entries for the selected period
Total Expenses	Sum of all expense entries for the selected period
Net Savings	Income minus Expenses — the bottom line of financial health
Savings Rate %	Percentage of income successfully saved
MoM Trends	Month-over-Month change in spending to detect seasonal patterns

2. Data Architecture — Star Schema

To enable efficient filtering, slicing, and time-intelligence calculations in Power BI, the raw transaction data is structured into a Star Schema — the industry-standard data modelling approach for analytical dashboards.

2.1 What is a Star Schema?

A Star Schema consists of one central Fact Table surrounded by multiple Dimension Tables. The Fact Table stores numerical transaction data. Dimension Tables store descriptive attributes used for filtering and grouping.

2.2 Fact Table

Table Name: fact_transactions — The central ledger of all financial entries
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Column	Description
Date	The date of the transaction (used for time-based analysis)
Account_ID	Foreign key linking to dim_accounts
Category_ID	Foreign key linking to dim_categories
Type	Classifies the entry as either Income or Expense
Amount	The monetary value of the transaction (always positive)

2.3 Dimension Tables

dim_accounts — Payment Methods

- Credit Cards (e.g., Chase, HDFC)
- Bank Accounts (Savings, Checking)
- Cash transactions

dim_categories — Hierarchical Spending Classification

Category	Sub-categories
Food	Groceries, Restaurants, Food Delivery
Housing	Rent, Electricity, Internet, Water
Transport	Fuel, Metro, Cab, Car Maintenance
Entertainment	OTT Subscriptions, Movies, Events
Healthcare	Medicine, Doctor Visits, Insurance
Income	Salary, Freelance, Other Income

dim_date — Calendar Table

- Generated using CALENDAR() or CALENDARAUTO() DAX function
 - Contains: Date, Day, Month, Month Name, Quarter, Year, Week Number
 - Marked as a Date Table in Power BI for accurate time intelligence
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3. Data Cleaning & ETL (Power Query)

Before any visualization, raw data from Excel is loaded into Power Query Editor for cleaning and transformation. This ETL (Extract, Transform, Load) process ensures data accuracy and consistency throughout the dashboard.

Step 1 — Data Type Formatting

- Amount column set to Fixed Decimal Number to prevent floating-point errors
- Date column enforced as Date type with a consistent format (YYYY-MM-DD)
- Type column (Income/Expense) confirmed as Text

Step 2 — Sign Standardization

- All transaction amounts stored as positive values in the fact table
- The Type column (Income/Expense) drives the logic — no negative signs needed in raw data
- DAX measures handle the arithmetic when calculating Net Savings

Step 3 — Handling Blank Sub-Categories

- Empty sub-category cells filled with the string 'Other' to prevent null values
- Ensures every transaction appears in the dashboard — no records dropped from filters

Step 4 — Removing Duplicates & Errors

- Duplicate transaction rows identified and removed
- Rows with null amounts or missing dates flagged and handled
- Category_ID and Account_ID validated against their dimension tables

4. DAX Measures — The Analysis Engine

All analytical calculations are written as DAX (Data Analysis Expressions) measures, stored in a dedicated Measures Table. This keeps the data model clean and the measures reusable across all visuals.

4.1 Core Measures

Total Income

```
Total Income = CALCULATE(SUM(fact_transactions[Amount]),  
    fact_transactions[Type] = "Income")
```

Filters the fact table to only Income rows and sums the Amount column.

Total Expenses

```
Total Expenses = CALCULATE(SUM(fact_transactions[Amount]),  
    fact_transactions[Type] = "Expense")
```

Filters the fact table to only Expense rows and sums the Amount column.

Net Savings

$\text{Net Savings} = [\text{Total Income}] - [\text{Total Expenses}]$

Reuses the two measures above to calculate the bottom line.

Savings Rate %

$\text{Savings Rate} = \text{DIVIDE}([\text{Net Savings}], [\text{Total Income}], 0)$

Uses DIVIDE() to safely handle cases where Total Income is 0, returning 0 instead of an error.

4.2 Time Intelligence Measures

Previous Month Expenses

$\text{PM Expenses} = \text{CALCULATE}([\text{Total Expenses}], \text{DATEADD}(\text{dim_date}[\text{Date}], -1, \text{MONTH}))$

Month-over-Month Change %

$\text{MoM Change \%} = \text{DIVIDE}([\text{Total Expenses}] - [\text{PM Expenses}], [\text{PM Expenses}], 0)$

Enables the trend chart to show whether spending increased or decreased versus the previous month.

5. Dashboard Design & Visuals

The dashboard layout is designed for immediate readability — a user should understand their financial health within seconds of opening the report.

5.1 Dashboard Screenshot

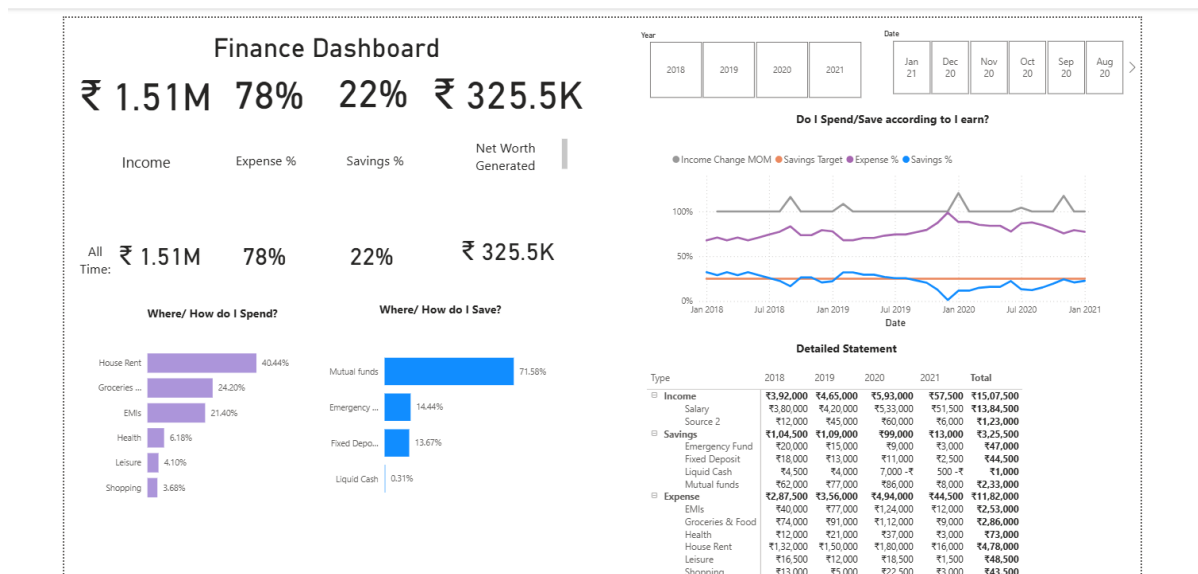


Figure 1: Personal Finance Dashboard — Power BI (2018–2021)

5.2 KPI Header Block

Four prominent KPI cards across the top display the most critical metrics at a glance:

KPI Card	Value Shown / Purpose
Total Income (INR)	Rs. 1.51M — total earnings across all time periods
Expense %	78% — proportion of income spent
Savings %	22% — proportion of income saved (the health indicator)
Net Worth Generated	Rs. 325.5K — cumulative wealth built over the period

5.3 Core Visuals

Do I Spend/Save According to What I Earn? (Line Chart)

- Tracks Income Change MoM, Savings Target, Expense %, and Savings % over time
- Multi-line chart spanning Jan 2018 to Jan 2021 — reveals spending vs saving trends
- Instantly shows months where expenses spiked above the savings target line

Where / How Do I Spend? (Horizontal Bar Chart)

- House Rent: 40.44% — the single largest expense category
- Groceries: 24.20% | EMIs: 21.40% — the next two biggest drains
- Health, Leisure, Shopping make up the remaining ~14%

Where / How Do I Save? (Horizontal Bar Chart)

- Mutual Funds: 71.58% — primary savings vehicle
- Emergency Fund: 14.44% | Fixed Deposit: 13.67%
- Liquid Cash: only 0.31% — minimal idle cash, most money is invested

Detailed Statement (Matrix Table)

- Year-wise breakdown of Income, Savings, and Expenses from 2018 to 2021
- Drill-down to sub-categories: Salary, Groceries & Food, EMIs, House Rent etc.
- Cross-filters dynamically with all other visuals on the page

5.4 Slicers & Filters

- Year slicer: 2018, 2019, 2020, 2021 — for year-wise analysis
 - Date slicer: Monthly granularity with scrollable month buttons
 - All visuals respond to slicer selections simultaneously
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6. Key Insights from the Dashboard

Based on the data visible in the dashboard (2018–2021):

- House Rent alone consumed 40.44% of all expenses — a clear financial pressure point
- Despite high expense %, a 22% savings rate was maintained consistently
- 71.58% of savings went into Mutual Funds — showing investment-first mindset
- Net worth of Rs. 3.25 Lakh was generated over the 3-year period
- Income remained relatively stable (MoM line flat) while expenses fluctuated — visible in the trend chart

7. Tools & Technologies Used

Tool / Technology	Purpose in This Project
Power BI Desktop	Primary tool for data modelling, DAX, and dashboard creation
Power Query (M Language)	ETL — data cleaning and transformation before loading
DAX	Writing measures for KPIs, aggregations, and time intelligence
Microsoft Excel	Source data — transaction logs maintained in Excel sheets
Star Schema Design	Data modelling approach for optimized performance and filtering
GitHub	Version control and portfolio documentation

— End of Documentation —

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